SELECTED ENGINEERING WORK

PROJECT PORTFOLIO

Mechatronics • Robotics • Embedded Systems

YESHWANTH S

B.Tech Mechatronics Engineering

SRM Institute of Science and Technology

ROBOTICS • SENSING • AUTOMOTIVE SYSTEMS

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01 / PROFILE

Engineering interests grounded in practical systems

I am a B.Tech Mechatronics Engineering student at SRM Institute of Science and Technology. My coursework includes embedded systems, control systems, electric vehicle technology, Python programming, localization and state estimation, perception for autonomous systems, and robotics. My projects explore autonomous navigation, sensor-based prototypes, embedded tracking, and robotic mechanisms.

7.9	5 projects	2 internships

TECHNICAL AREAS

ROS 2 · Python · OpenCV · CARLA Simulator · Arduino · ESP32 · GPS · Ultrasonic sensors · LiDAR · Robotics · Vehicle inspection · Automotive production and quality inspection

FEATURED PROJECTS

Pick-and-Place Robot Rover: mechanical function and movement control; original files unavailable, so reconstructed documentation is clearly labelled.

Vehicle Speed Detection: ultrasonic sensing and measurement logic, with new example code.

Differential Drive Navigation: LiDAR-based navigation concept with an illustrative path demo.

Supporting projects: ROS 2 Laneless Navigation and ESP32 GPS Tracker, each with newly created demonstration material.

EVIDENCE TRANSPARENCY

The rover was previously built according to the author's recollection, but original photos and code are unavailable. Code examples and diagrams in this package are newly created and are not represented as original project records. Simulations are illustrative unless explicitly run and tested by the author.

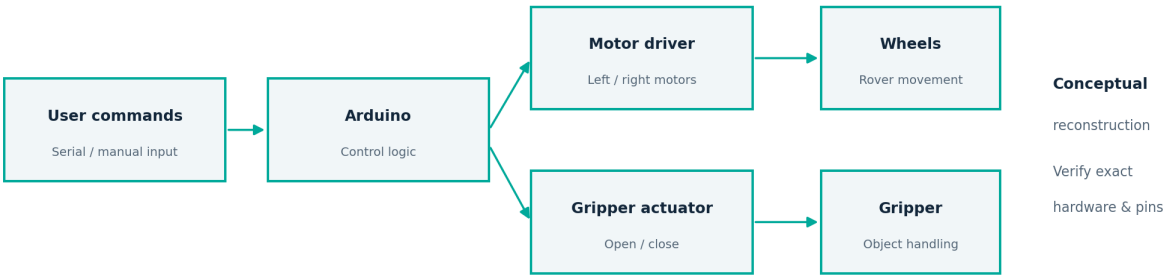
02 / FEATURED BUILD

Autonomous Pick-and-Place Robot Rover

Arduino • Wheels • Gripper • Robotics

A previously built rover combining wheeled movement with a gripper for picking up and placing objects. The exact original controller model, motor driver, actuator type, wiring, and test results are not available.

CONCEPTUAL SYSTEM DIAGRAM



WHAT THIS PROJECT EXPLORES

- Coordinating rover motion and gripper actions.
- Integrating a controller with motor-drive electronics and a mechanical end-effector.
- Positioning the rover before gripping and releasing an object.
- Mechanical stability and repeatability of the pick-and-place sequence.

NEW DEMONSTRATION CODE

A new Arduino serial-control example is included in the source package. It supports manual movement commands and servo gripper open/close commands under stated hardware assumptions. It is not recovered original code and must be adapted to the real wiring before use.

DOCUMENTATION STATUS

Diagram: reconstructed concept. Starter code: newly authored, not hardware-tested. Original photos: unavailable. Measured results: not available. To strengthen this project, run the code safely on compatible hardware, photograph the setup, and record actual pick-and-place trials.

03 / SENSOR & EMBEDDED PROJECTS

Vehicle Speed Detection System

Arduino • Ultrasonic sensors

The resume describes a prototype using ultrasonic sensors and Arduino to estimate and display vehicle speed. A new two-sensor timing example is provided as a learning demonstration; sensor placement and detection behavior require physical calibration and testing.

GPS Tracker using ESP32

The resume describes GPS data acquisition and wireless communication for location tracking. A new ESP32 UART example reads location data from a compatible GPS module and prints coordinates to the serial monitor. It requires the TinyGPSPlus library and correct wiring.

SOURCE FILES

- source_code/vehicle_speed_detection/vehicle_speed_two_sensor_demo.ino
- source_code/esp32_gps_tracker/esp32_gps_tracker_demo.ino

VALIDATION TO RECORD

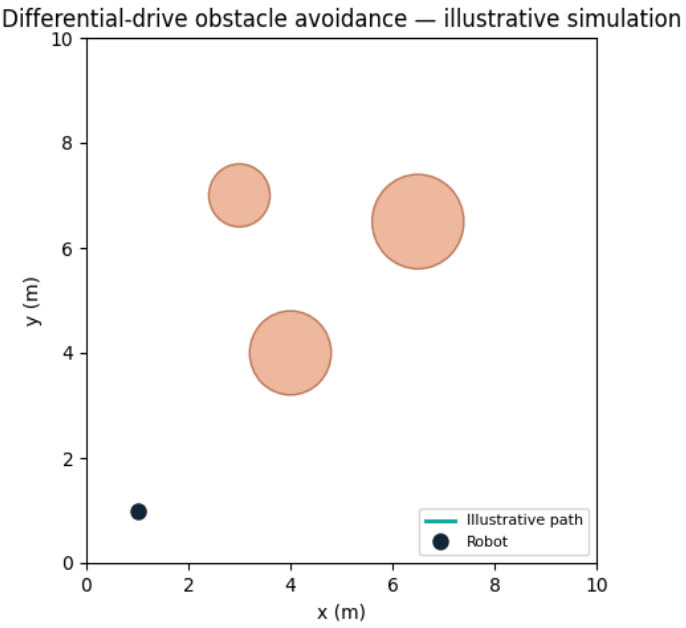
For the speed detector, measure actual sensor spacing and compare estimated speed with a known reference over repeated trials. For the GPS tracker, verify satellite fix outdoors and compare reported coordinates with a trusted map/location reference. Do not report accuracy until tests have been performed.

04 / ROBOTICS & AUTONOMY

Differential Drive Robot Navigation

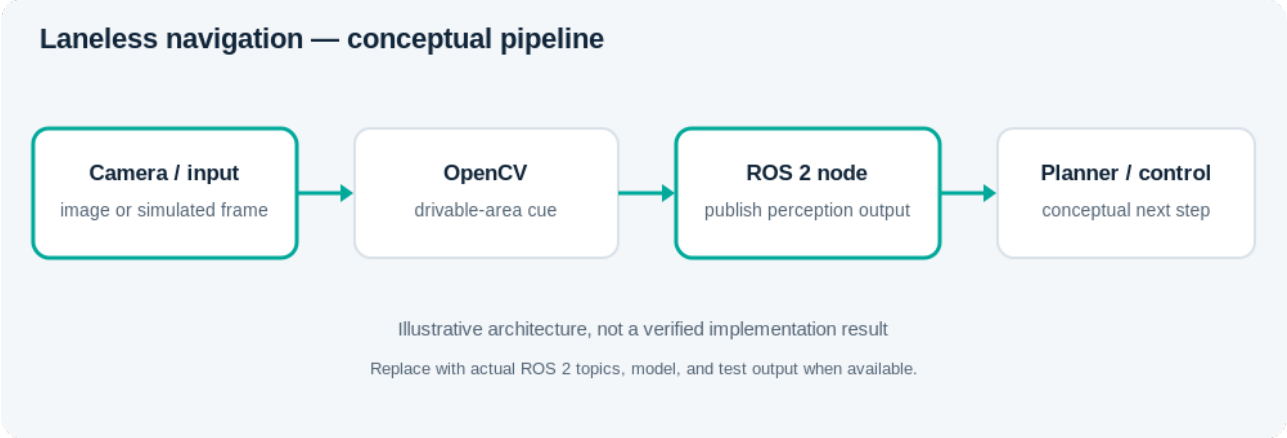
ROS 2 • Python • LiDAR

The resume describes a ROS 2 and Python navigation system integrating LiDAR-based obstacle detection and avoidance. The package includes an illustrative plotted route around obstacles; it is a path illustration, not a physics simulator or a verified ROS 2 run.



ROS 2 Laneless Navigation

The resume describes a ROS 2/CARLA concept using camera and simulated sensor data with Python/OpenCV for drivable-area detection. A separate synthetic OpenCV demonstration is included to explain image masking. It is not the original project, not CARLA, and not a validated road detector.



SOURCE FILES

- source_code/differential_drive_demo/differential_drive_demo.py
- source_code/laneless_navigation_demo/laneless_opencv_demo.py

05 / EXPERIENCE & EDUCATION

Automotive experience

Yamaha Authorized Service Center – Maruthi Motors

Service Center Intern · Tirupathur, Tamil Nadu · Jun–Jul 2026

Troubleshot engine, brake, suspension, and electrical-system faults; performed inspections, routine maintenance, servicing, and basic repairs on 15+ two-wheelers.

Renault Nissan Automotive India Pvt. Ltd.

Shop Floor Intern · Chennai, Tamil Nadu · Dec 2025–Jan 2026

Supported Trim and Chassis assembly-line operations, process monitoring, quality inspection, defect identification, and quality-improvement initiatives aimed at reducing DPHU.

ACHIEVEMENTS & LEADERSHIP

- 1st Place — Drone Racing, IITDM.
- 2nd Place — Sky Sprint, Tantratsav 2026, Amrita Vishwa Vidyapeetham, Chennai.
- 2nd Place — Touchdown-X, Tantratsav 2026, Amrita Vishwa Vidyapeetham, Chennai.
- Committee Head, Robogyan, Aaruush, SRM — led a 20-member team.
- Event Coordinator, RoboSoccer; Organizing Team Member, ICARM 2026.

EDUCATION & CONTACT

B.Tech Mechatronics Engineering — SRM Institute of Science and Technology · 2023–Present · CGPA 7.9

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HOW TO USE THE SOURCE PACKAGE

The ZIP package includes a website portfolio, diagrams, a demo GIF, source files, and README instructions. Python demos require dependencies listed in the README. Hardware sketches must be checked against the actual board, motor driver, voltage, and actuator before powering any circuit.